EE610: Analog IC design (2025)

Design project #2

Input Device	Roll No.	R_L	C_L
NMOS	251040079	20kΩ	1pF

NAME : PURWANSH SAHU

ROLL NUMBER : 251040079

- **Schematic of two-stage Opamp with CMFB :**

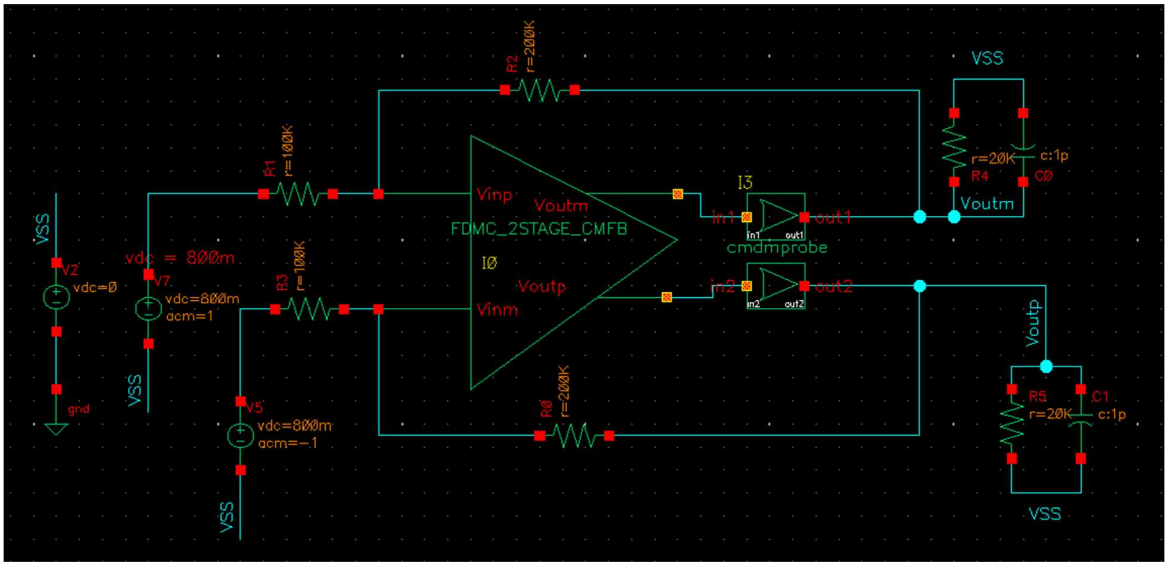


Figure 1: Fully-differential two-stage Miller opamp in -2 inverting configuration

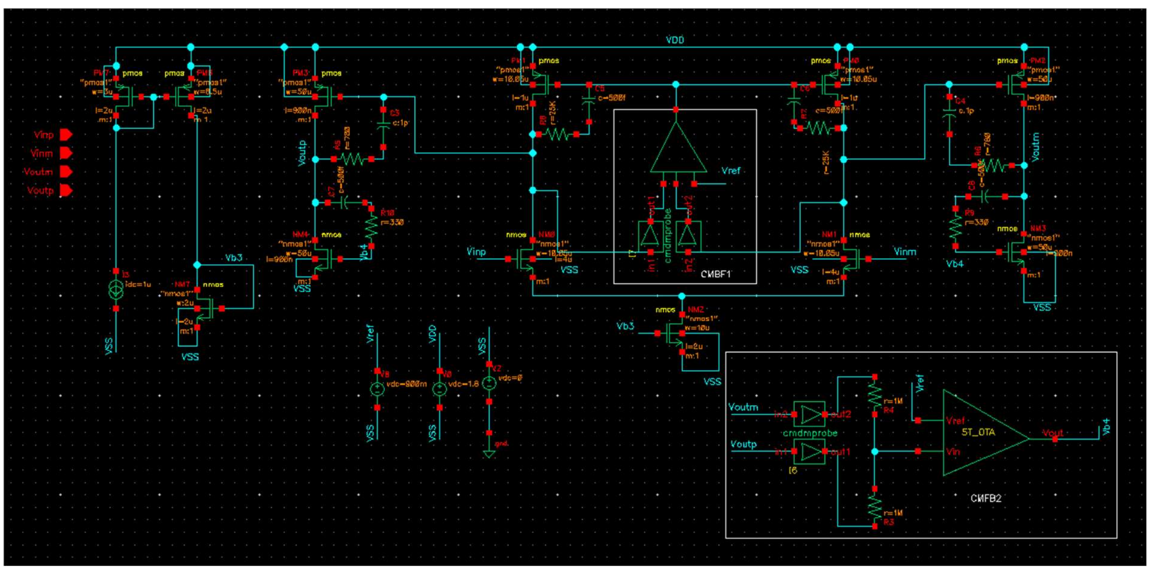
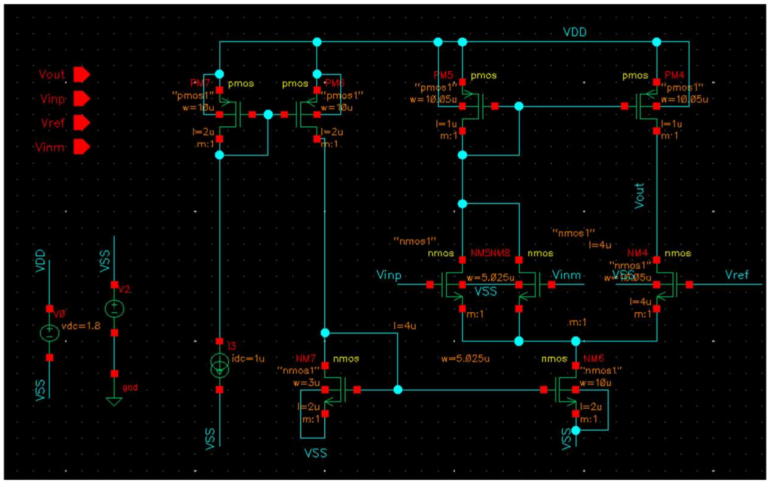
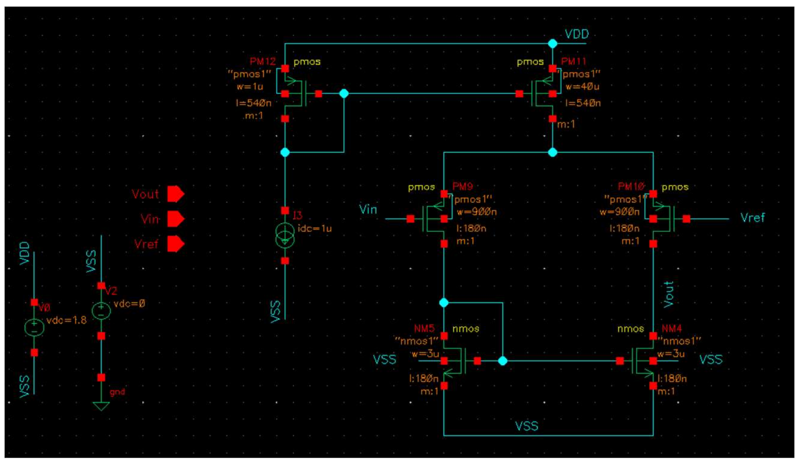


Figure 2: Transistor level fully-differential two-stage Miller opamp with CMFB circuits



(a) CMFB1 : nMOS input 5-transistor OTA for first stage common-mode control



(b) CMFB2 : pMOS input 5-transistor OTA for second stage common-mode control

1. Specifications Analysis

The operational amplifier is designed for a closed-loop gain of −2, using R_1 = 100 kΩ and R_2 = 200 kΩ (feedback factor β = 1/3). The design must achieve a DC loop gain of at least 60 dB, while driving a load of RL = 20 kΩ ∥ CL = 1 pF.

Key performance requirements include:

- A closed-loop 3-dB bandwidth ≥ 1 MHz without peaking,
- CMFB loops achieving ≥ 60° phase margin,
- CMFB unity-gain bandwidths of at least one-quarter of the differential UGB,
- Second-stage common-mode output set at V_{CM} = VDD/2,
- All CMFB resistor values kept 1 MΩ or below.

2. Architecture Description:

The operational amplifier employs a fully differential two-stage structure with Miller compensation and dedicated Common-Mode Feedback (CMFB) loops.

- **First Stage:**
  nMOS differential pair (NM1, NM0) with pMOS active loads (PM0, PM1).

- **Second Stage:**
  pMOS differential pair (PM2, PM3) with nMOS active loads (NM3, NM4).
- **Bias Network:**
  All bias currents are supplied through current mirrors referenced to a 1 µA source.
- **CMFB Implementation:**
  Two CMFB loops—one for each stage—using a 5-transistor OTA configuration.
- **Compensation Strategy:**
  Miller capacitors with series resistors are applied across the differential path and the CMFB loops for stable frequency compensation.

3. Design Methodology

Specifications Analysis:

The op-amp is designed for a closed-loop gain of −2 using R_1 = 100 kΩ and R_2 = 200 kΩ (β = 1/3).

It must deliver a minimum DC loop gain of 60 dB while driving **R_L = 20 kΩ ∥ C_L = 1 pF**.

Key requirements:

- Closed-loop 3-dB bandwidth ≥ **1 MHz** with no peaking
- CMFB phase margin ≥ **60°**
- CMFB UGB ≥ **¼ of differential UGB**
- Second-stage output common-mode fixed at **VCM = VDD/2**
- All CMFB resistors ≤ **1 MΩ**

- **Gain Distribution**

The loop-gain requirement is:

$$20\log\left(A_1 \times A_2 \times \beta\right) \geq 60 \text{ dB}$$

which simplifies to:

$$A_1 + A_2 \geq 60 - 20\log\left(1/3\right) \approx 69.54 \text{ dB}$$

Selected gain allocation:

- First-stage gain: ~46 dB
- Second-stage gain: ≥ 24 dB

First Stage Design

The target first-stage gain is

$$A_1 = g_m(r_{on} \parallel r_{op}) \approx 315,$$

with $r_{on} = r_{op} = r_o$, leading to

$$g_m r_o \approx 630.$$

NM0–NM1 (Input Differential Pair):

Based on the g_m/g_{ds} characteristics:

- $L = 4\,\mu$m
- $V_{OV} = 102$ mV

- $g_m/g_{ds} = 630$
- $I_D = 7.1\ \mu\text{A}$
- $I_D/W = 0.711\ \mu\text{A}/\mu\text{m} \rightarrow W = 10.05\ \mu\text{m}$
- $g_m/I_D = 12.91 \rightarrow g_m = 91.6\ \mu\text{S}$

PM0–PM1 (pMOS Active Loads):

From the I_D/W graph:

- $I_D = 7.1\ \mu\text{A}$
- $I_D/W = 0.706 \rightarrow W = 10.05\ \mu\text{m}$
- $L = 1\ \mu\text{m}$, $V_{OV} = 132.6\ \text{mV}$

NM2 (Tail Current Source):

- $I_D = 14.2\ \mu\text{A}$
- $I_D/W = 1.421 \rightarrow W = 10\ \mu\text{m}$
- $L = 2\ \mu\text{m}$, $V_{OV} = 98\ \text{mV}$

Second Stage Design

The second-stage gain requirement is:

$$A_2 = g_m(r_{on} \parallel r_{op} \parallel R_L) \geq 18,$$

with $R_L = 20\ \text{k}\Omega$ and $r_o \approx 120\ \text{k}\Omega$.

This gives:

- $g_m \geq 1.2\ \text{mS}$
- $g_{ds} \approx 8.33\ \mu\text{S}$

PM2–PM3 (Second-Stage Input Pair):

From device curves:

- $L = 900\ \text{nm}$
- $V_{OV} = 440\ \text{mV}$
- $g_m/g_{ds} = 144$
- $g_{ds}/I_D = 0.064404 \rightarrow I_D = 320\ \mu\text{A}$
- $I_D/W = 6.4 \rightarrow W = 50\ \mu\text{m}$

NM20–NM21 (Active Loads):

- $g_{ds}/I_D = 0.04$
- $L = 900\ \text{nm}$, $V_{OV} = 127\ \text{mV}$
- $I_D/W = 5.44 \rightarrow W = 50\ \mu\text{m}$

Compensation Network Design

CMFB1 Stability (First Stage CMFB)

Stability requirement:

$$R_{7,8} \geq \frac{1}{2g_m(PM1, PM0)} = \frac{1}{2 \times 90.2\ \mu\text{S}} = 5.54\ \text{k}\Omega.$$

Chosen values:

- $R_6, R_7 = 25\ \text{k}\Omega$

Simulation-based tuning:

- $C_5, C_6 = 500\ \text{fF} \rightarrow$ Phase Margin $PM = 61.12°$

CMFB2 Stability (Second Stage CMFB)
Stability requirement:

$$R_{8,9} \geq \frac{1}{2g_m(NM3, NM4)} = \frac{1}{2 \times 3.018\ \text{mS}} = 330\ \Omega.$$

Chosen values:

- $R_8, R_9 = 330\ \Omega$

Simulation tuning:

- $C_7, C_8 = 500$ fF→ Phase Margin $PM = 72.59°$

Differential Loop Compensation
Stability requirement:

$$R_{10,11} \geq \frac{1}{2g_m(PM2, PM3)} = \frac{1}{2 \times 1.28\ \text{mS}} = 390\ \Omega.$$

Chosen values:

- $R_{10}, R_{11} = 780\ \Omega$p

Simulation tuning:

- $C_3, C_4 = 1$ pF→ Phase Margin $PM = 64.05°$

Block-level Design Summary

Parameter	First Stage	Second Stage	Main Diff. Loop	CMFB1	CMFB2
G_m	90.2µS	1.28mS	-	-	-
I_D	13.53µA	318µA	-	-	-
C_C	-	-	1pF	500fF	500fF
R_C	-	-	780Ω	25kΩ	330Ω
PM	-	-	64.05°	61.12°	72.59°
UGB	-	-	3.95 MHz	4.37 MHz	11.4 MHz

- **Result #1: Simulated DC Operating Point**

Parameter	Designed	Simulated	Units
First Stage g_m	91.6	90.2	µS
First Stage I_D	14.2	13.53	µA
Second Stage g_m	1200	1281.67	µS
Second Stage I_D	320	318	µA

Table1: DC Operating point results

- **Result #2: Differential Mode Loop Gain**

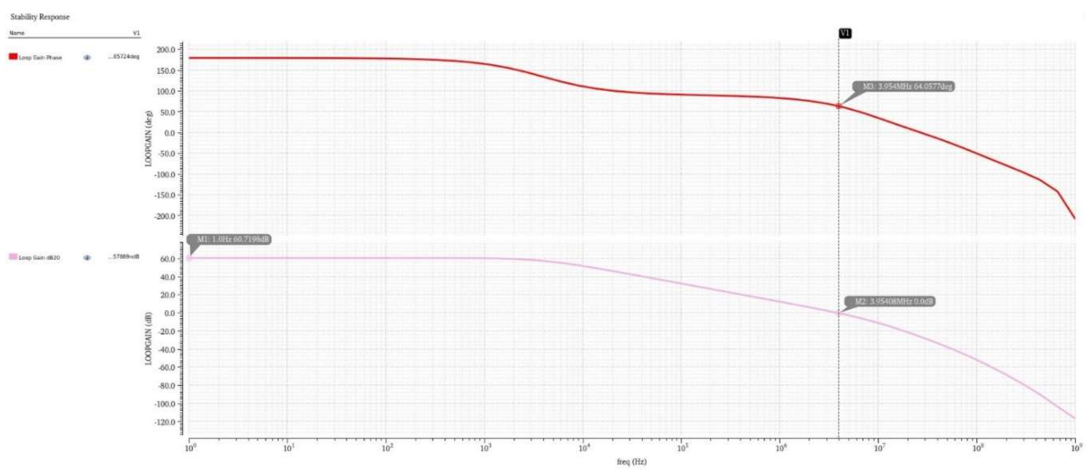


Figure 3: Differential-mode loop gain mag and phase response

Loop Gain	UGB	Phase Margin
60.72 dB	3.95 MHz	64.05 deg

- **Result #3: Closed-loop Frequency Response**

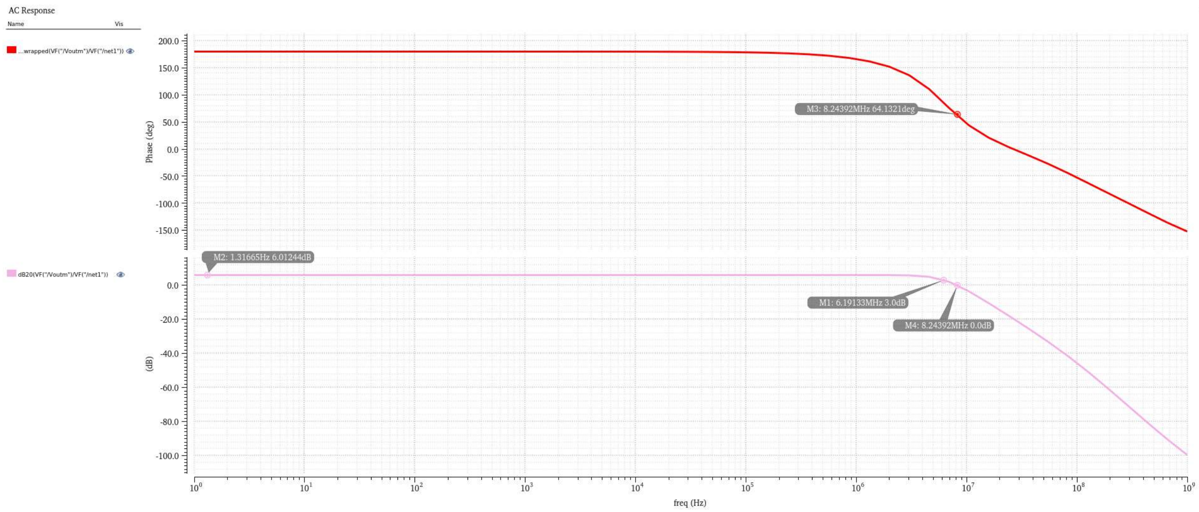


Figure 4: Closed-loop frequency response

Close loop Gain	-3dB Bandwidth	UGB	Phase Margin
6 dB	6.19 MHz	8.24 MHz	64.13 deg

- **Result #4: CMFB Loop Gains**

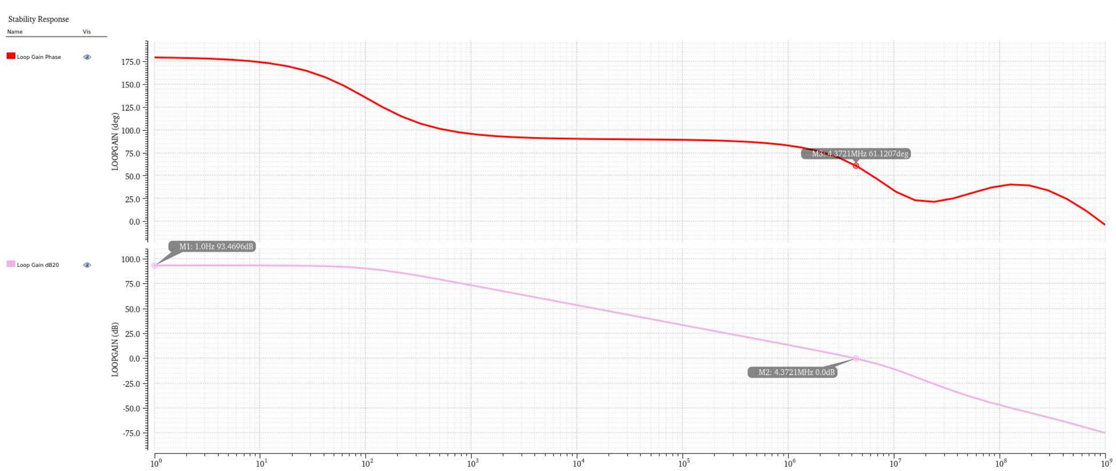


Figure 5: CMFB1 loop gain magnitude and phase response

Loop Gain	UGB	Phase Margin
93.47 dB	4.37 MHz	61.12 deg

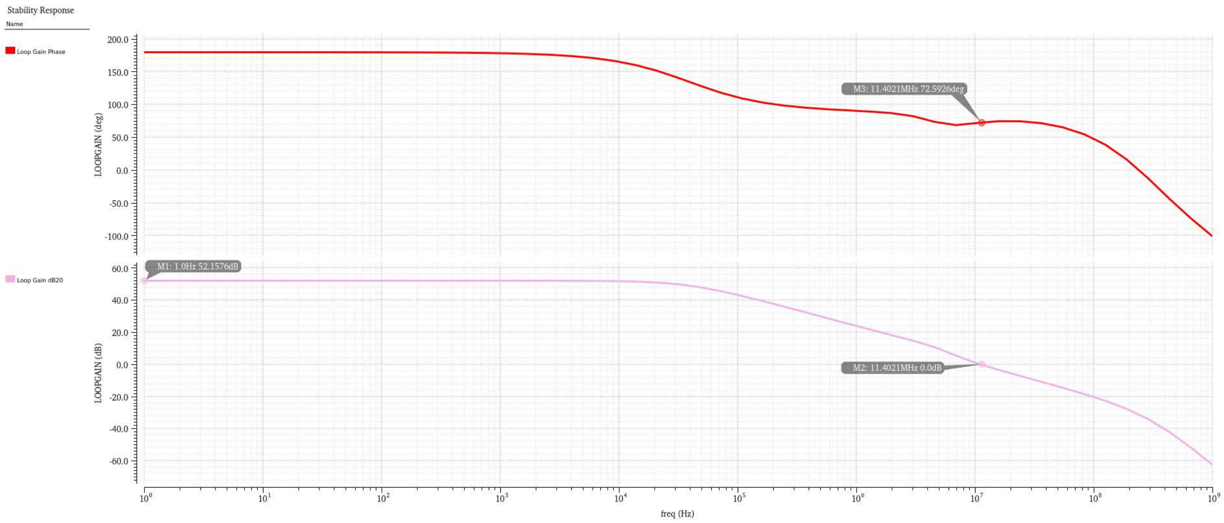


Figure 6: CMFB2 loop gain magnitude and phase response

Loop Gain	UGB	Phase Margin
52.16 dB	11.4 MHz	72.59 deg

- **Result #5: Differential Step Response**

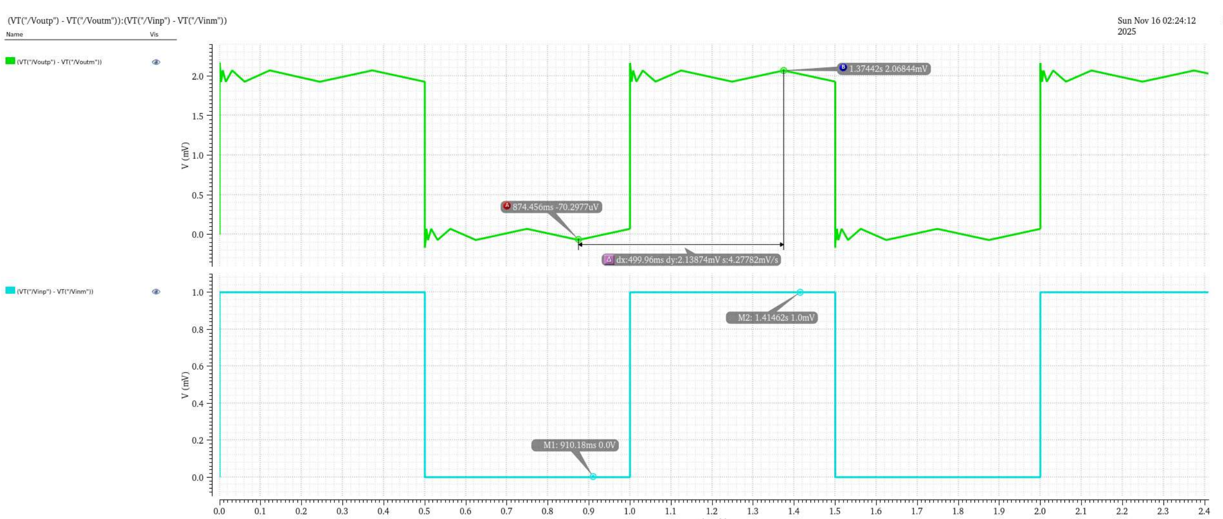


Figure 7: Differential output response for 1mV differential step input

- **Result #6: Common-mode Step Response**

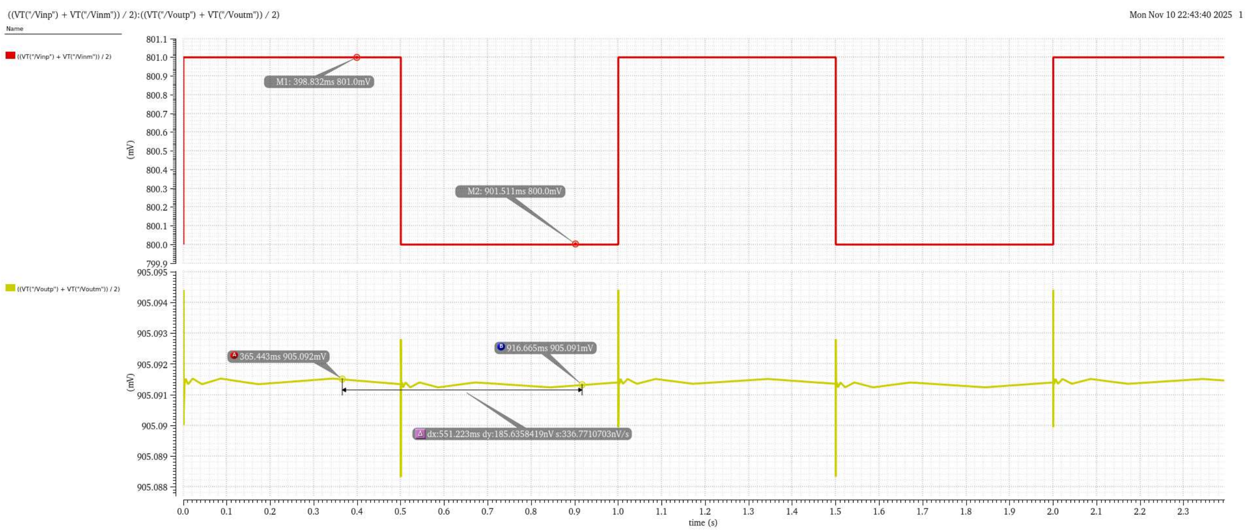


Figure 8: Output response for 1mV common-mode step input

Explanation: The 1 mV differential step produces a 2.06 mV differential output, which closely matches the expected closed-loop differential gain of $A_{DM} \approx 2$, set by the feedback network (R2/R1 = 200k/100k = 2). This confirms that the amplifier operates correctly in its intended closed-loop configuration.

In contrast, when a 1 mV common-mode step is applied to both inputs, the differential output changes by only about 185 nV, indicating that the common-mode gain is effectively $A_{CM} \approx 0$. This behaviour is expected because the tail current source maintains nearly constant current, causing both sides of the differential pair to experience identical variations. As a result, common-mode disturbances do not translate into differential output changes, demonstrating strong common-mode rejection and proper CMFB operation.